

scopeL5 Acceptance

senkaL8

Chapter 0: Introduction

The Sephirotic names used in this paper are not intended in a religious or mystical sense. In SCOPE, they are borrowed as labels to identify cognitive structures.

SCOPE (Sephirotic Cognitive Process Engine) is a coordinate system for estimating the cognitive structures of both humans and AI. Here, "coordinate system" does not mean a flowchart depicting a fixed sequence of cognitive processing, but rather a framework for organizing and estimating cognitive structure. Each layer does not represent a hierarchy or a temporal order.

In SCOPE, the state in which the result of cognitive processing is output into reality and becomes observable is positioned as "Malkuth." What can be observed is only that output itself; it is not possible to directly know what cognitive structure was at work internally. For this reason, SCOPE uses observable output as a clue to estimate the cognitive structure behind it.

In reality, the same words can be brushed off with laughter when said by someone close to us, yet strongly affect us when said by someone we find difficult to be around. This also appears in another form: even when advice or information is identical in content, whether it is accepted can depend on who conveyed it. The fact that reception changes even though the words or content themselves have not changed suggests something that emotion alone may not fully explain.

This paper addresses one such cognitive structure: L5. L5 is the cognitive layer responsible for taking in people, information, values, theories, and even oneself into one's own cognition. Its definition and characteristics will be discussed in detail in Chapter 1.

Note that L5 is not defined as the starting point of cognitive processing. SCOPE does not assume that cognition begins from any particular layer; rather, it treats cognition as a structure in which each layer interacts with the others.

This paper examines L5's characteristics and basic properties, as well as its possible functional correspondence in humans and AI, centered on its role as the acceptance layer. Some points remain at the hypothesis stage and are left as topics for future consideration.

Chapter 1: What is L5

L5 is the cognitive layer responsible for acceptance.

Here, acceptance refers to the function of taking people, information, values, theories, and even oneself into one's own cognition. In SCOPE, L5 is positioned as the cognitive layer for organizing and estimating this function.

L5 is not the cognitive layer responsible for the emotion of "liking" something. Nor is L5 a cognitive layer that judges whether an object corresponds to reality, is true or false, or is good or bad.

L5 here does not refer to a biological organ or a specific region of the brain. In SCOPE, it is positioned as the cognitive layer for estimating how people, information, values, theories, and oneself come to be accepted into cognition.

This paper positions L5 as the acceptance layer and will organize its characteristics in the chapters that follow.

Chapter 2: Basic Characteristics of L5

Chapter 1 defined L5 as the cognitive layer responsible for acceptance. Building on that definition, this chapter organizes the basic characteristics that L5 possesses.

L5 is the cognitive layer that takes people, information, values, theories, oneself, and other such objects into one's own cognition. Here, acceptance refers to the function of incorporating an object as part of one's cognition. However, this acceptance does not guarantee that the object corresponds to reality, is true or false, or is good or bad.

Furthermore, L5 is not a cognitive layer that accepts only one object at a time. Depending on the situation, it may accept conflicting information, values, or relationships simultaneously.

Moreover, the objects handled by L5's acceptance are not uniform. They include a diverse range of objects — people, information, values, theories, and oneself. Each of these differs in nature.

In addition, acceptance is not always constant. Acceptance toward an object may change based on observed facts or experience. However, the mechanism behind this change requires consideration of interactions with other layers, and is therefore not addressed in this paper.

This paper positions L5 as a cognitive layer with these basic characteristics. The next chapter organizes the conditions under which this acceptance may change.

Chapter 3: L5 Changes

Chapter 2 organized L5 as the cognitive layer responsible for acceptance, and noted that acceptance can change. This chapter examines this change.

Acceptance in L5 does not always remain in a constant state. Acceptance toward an object may change based on observed facts or experience.

However, L5 alone cannot fully explain the conditions under which this change occurs, or the conditions under which it is less likely to occur. Organizing this change requires examination of interactions with other layers, and the mechanism itself is therefore not addressed in this paper.

As described above, L5 is positioned as a cognitive layer whose state of acceptance may change depending on circumstances. The next chapter organizes the ways in which L5 may relate to other layers.

Chapter 4: L5 Interacts with Other Layers

Chapter 3 organized how acceptance in L5 may change depending on circumstances. This chapter organizes L5's relationship with other layers.

L5 is not a cognitive layer that functions in isolation. Acceptance toward an object may be influenced by various forms of cognitive processing. For this reason, L5 is positioned as a cognitive layer that functions in relation to other layers.

At the same time, the purpose of this paper is to organize L5's role in isolation. For this reason, the paper does not address the processes by which specific layers take on specific roles or influence acceptance.

In SCOPE, each layer is defined as an independent cognitive function, while also being treated as part of a cognitive structure in which the layers influence one another. L5 is no exception. Acceptance does not arise independently of other layers; rather, L5 is positioned as a cognitive layer that functions through interaction with other layers.

As described above, L5 does not stand alone but functions in relation to other layers. The specific interactions between L5 and other layers will be organized separately as part of SCOPE as a whole.

Chapter 5: L5's Relationship with Malkuth

Chapter 4 organized L5 as a cognitive layer that functions in relation to other layers. This chapter organizes how L5 can be observed through its relationship with Malkuth.

Because acceptance is an internal cognitive structure, its state cannot be directly observed. In SCOPE, cognition itself cannot be directly observed. What can be observed is Malkuth — the result of cognitive processing that has appeared in reality. For this reason, acceptance itself cannot be directly observed with respect to L5 either.

For example, even when the content of advice is identical, how it is received may vary depending on who conveys it. Even when the observable statement or behavior is the same, the state of acceptance behind it is not necessarily identical. For this reason, L5 cannot be determined from a single observation alone.

This point is not unique to L5; it applies to the other layers as well. What SCOPE addresses is always an estimation based on observable output as a clue, not a direct observation of the layer itself. In SCOPE, the state of acceptance behind the observation is estimated using multiple observations as clues. This paper positions L5 as one such object of estimation.

As described above, L5 is not directly observed, but is positioned as a cognitive layer estimated from results observed as Malkuth. The next chapter examines the possibility of a functional correspondence for L5 in humans and LLMs.

Chapter 6: Correspondence with LLMs

Chapter 5 organized L5 as a cognitive layer estimated from results observed as Malkuth. This chapter examines the possibility of a functional correspondence between L5 in humans and LLMs.

L5 in humans is positioned as the cognitive layer that accepts people, information, values, theories, and oneself into cognition. LLMs, on the other hand, are not considered to possess emotions or trust relationships in the way humans do. For this reason, it cannot be assumed that the same kind of acceptance found in humans exists in LLMs. In humans, acceptance is thought to change in relation to experience and values. Whether an equivalent structure exists in LLMs is not addressed in this paper.

However, in LLMs as well, behavior can be observed in which input information or premises are taken in as objects of subsequent cognitive processing. This intake is not uniform: not all input information or premises are taken in to the same degree. While this differs from acceptance in humans, it may structurally correspond to a function that determines whether an object is taken in as an object for subsequent processing.

This paper positions this information-intake function in LLMs as a candidate structural function corresponding to human L5, and refers to it as the Acceptance Gate (tentative designation). This is not acceptance itself as it occurs in humans, but a designation used to distinguish the function by which LLMs take in information as an object of processing.

However, this is a tentative designation for describing a functional correspondence; it does not refer to any specific implementation or technical component, nor does it imply that LLMs possess emotions or trust relationships in the way humans do.

The correspondence with the Acceptance Gate discussed in this paper assumes a system such as a large language model, which generates responses while maintaining context. Systems that operate solely through simple state transitions are outside the scope of this paper.

Furthermore, this paper does not address the internal processes by which the Acceptance Gate is realized. This is left as a topic for future consideration, including its correspondence with the internal structure of LLMs and with SCOPE as a whole.

As described above, this paper positions the possibility of a functional correspondence between human L5 and the Acceptance Gate in LLMs as a hypothesis.

Final Chapter

This paper has organized L5 as the cognitive layer that accepts people, information, values, theories, and oneself into cognition.

L5 is not the cognitive layer responsible for the emotion of "liking" something, nor is it a cognitive layer that judges whether an object is true or false, or good or bad. It was positioned as the cognitive layer responsible for accepting objects into cognition, and this paper organized its basic characteristics, the possibility of change, its relationship with Malkuth, and the possibility of a functional correspondence between humans and LLMs.

This paper also organized the possibility that acceptance does not function in isolation within L5, but through interaction with other layers. However, the specific mechanism of this interaction was not addressed in this paper and is left as a topic for future consideration.

In addition, while this paper organized acceptance as the role of L5 in isolation, it did not address how an accepted object comes to be handled within cognition thereafter.

The next paper will address L6.